

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE REGION
FORM SIX PRE-MOCK EXAMINATION**

136/1

COMPUTER SCIENCE 1

(For Both School and Private Candidate)

Time 3 Hours

Tuesday, 25th August 2026 p.m

Instructions

1. This paper consists of sections A and B with a total of ten (10) questions.
2. Answer all questions in section A and two (2) questions from section B
3. Section A carries seventy (70) marks and section B carries thirty (30) marks.
4. All writing must be in blue or black ink, except drawings which must be in pencil.
5. Communication devices and any unauthorised materials are not allowed in the examination room.
6. Write your Examination Number on every page of your answer booklet(s).

SECTION A (70 Marks)

1. a) An automated industrial chemical reactor has an "**Emergency Relief Valve**" that only opens if specific conditions are met simultaneously.

Condition A: Reactor temperature exceeds 150°C.

Condition B: Internal pressure exceeds 5 atmospheres.

Condition C: The manual override safety switch is set to ACTIVE.

Required:

- (i) Draw the truth table for the above scenario. **(2 Marks)**

- (ii) If the temperature exceeds 150°C ($A=1$) and the safety switch is set to ACTIVE ($C=1$), but the internal pressure is normal ($B=0$), will the relief valve open? Why? **(2 Marks)**

- (iii) Draw a logic gate diagram representing this system. **(2 Marks)**

- b) Use laws of Boolean algebra to simplify the following expression. **(4 Marks)**

i. $ABC + \bar{A}BC + AB\bar{C} + \bar{A}\bar{B}C$

ii. $ABC + A\bar{B} + AB\bar{C}$

2. A company aims to analyze the performance of a new microprocessor prototype. During testing, they collect the following data from the memory hierarchy, The L1 data cache responds in 4 nanoseconds when it hits. Around 80% of all memory accesses are satisfied by the L1 cache. When an access misses L1, it goes to main memory, which takes 50 nanoseconds to deliver the requested data. The intern notices that some workloads (like image processing) experience noticeable delays, and they are considering whether to optimize cache performance or memory latency.

- (a) Based on the data above, calculate Average Memory Access Time (AMAT) and analyze the performance of this system. What does it suggest about the overall memory behavior? **(5 Marks)**
- (b) If software optimization could reduce the miss rate by half, discuss how that would likely impact system responsiveness and why. **(5 Marks)**

3. a) Explain four pillars of object-oriented programming (OOP) with one example each. **(4 marks).**
- b) Study the following C++ program, which is intended to calculate percentage of a score, then answer the questions that follow.

```
1  include <iostream>
2  using namespace std;
3
4  int main() {
5      int totalScore = 0;
6
7      double percentage = (85 / 100) * 100;
8
9      cout << "Percentage is: " percentage << endl;
10
11     if (totalScore = 0) {
12         cout << "No score recorded!" << endl;
13     }
14
15     return 0;
16 }
```

- i. Identify four errors in the program above. **(4 Marks)**
- ii. Rewrite the program with all the errors corrected. **(2 Marks)**
4. a) A file system directory index is represented internally as a binary tree. During a system recovery, the file system manager reconstructs the lost binary tree structure using two logged traversal sequences:
- Preorder Traversal:** 10, 20, 40, 80, 90, 50, 30, 60, 70
- Inorder Traversal:** 80, 40, 90, 20, 50, 10, 60, 30, 70
- i. Using the Preorder and Inorder traversal sequences provided, reconstruct and sketch the binary tree. **(3 marks)**
- ii. Identify the root node and the maximum depth (height) of the reconstructed tree. **(1 mark)**
- b) Describe the following operations performed on data structures: **(2 marks)**
- (i) Insertion

- (ii) Deletion
- (iii) Searching
- (iv) Traversing

c) Explain why data structures are important in computer programming.

Give two reasons? **(2 marks)**

d) Demonstrate how to declare a pointer in C++, and critically explain what happens in memory when the pointer is declared, including how it references or stores the address of a variable. **(2 marks)**

5. a) Apex Manufacturing Ltd is evaluating how to automate its production capabilities across three distinct business units:

Unit A: Produces a single model of aluminum soft drink cans at extremely high volumes (over 500,000 units per day) with zero variation in design.

Unit B: Manufactures custom furniture parts in seasonal batches (e.g., 500 units of Design X this week, 500 units of Design Y next week), where switching product lines takes several hours to retool equipment.

Unit C: Operates an advanced fulfillment facility where automated guided vehicles (AGVs), sorting arms, inventory tracking sensors, and central management software work interdependently in real time.

Questions:

(i) For Unit A and Unit B, identify the most appropriate type of automation system and explain how each type operates in relation to production volume, flexibility, and product variation.

(3 marks)

(ii) Name the type of automation implemented in Unit C. Explain one key advantage and one major vulnerability of deploying this specific architecture in Unit C's fulfillment facility. **(3 marks)**

(3 marks)

(iii) Describe five real-life applications of automated systems. **(4 marks)**

(4 marks)

6. a) Define an Operating System (OS) based on its structural placement within a computer system hierarchy. **(2 marks)**
- b) An Operating System is often compared to an orchestra's conductor, as it coordinates various components to work together harmoniously. Explain three functions Operating System **(3 marks)**
- c) A software development company, DevCore Tech, is engineering a real-time multiplayer gaming application designed to support thousands of concurrent users. During beta testing, developers noticed that when multiple applications run in the background, the game experiences severe frame rate drops, unexpected system crashes, and memory allocation delays. Additionally, third-party cheat software was able to modify game data directly in RAM because of weak memory isolation.
- i. Identify three key Operating System design principles that were compromised or inefficiently implemented in the scenario above. **(3 marks)**
- ii. Explain how the principle of Stability directly impacts user experience when running resource-intensive application software like DevCore's game. **(2 marks)**
7. a) At the very basic level, a computer system is a device consisting of three components. Mention and discuss these three components. **(3 Marks)**
- b) A Mr. Jones, a form five student at Mkwawa secondary school intends to buy a computer that he could use while at school as a tool while in the school as well in the hostels. Kindly advise him on the kind and type of a computer system he could buy. Give him the full specification of a good computer. **(4 Marks)**
- c) Explain the difference between the terms 'computer' and 'computer system'. **(3 Marks)**

SECTION B (30 Marks)

Answer two (2) questions in this section. Each question carries fifteen (15) marks.

8. Network model defines the protocols for network communications. Explain the functions of each open system interconnect model layer in network communication. **(15 Marks)**

9. As part of the government's "SmartNation Project," junior data analysts are being deployed across public and private industries to implement data-driven decision-making. Identify four sectors where data science is applied, and explain one key operational problem data science helps to solve within each sector. **(15 Marks)**

10. "ST. Jude Academy," a growing secondary school, currently tracks student admissions, fee payments, examination marks, and library book loans through separate desktop spreadsheets maintained by individual staff members. This setup has led to widespread problems, including duplicate student profiles, inconsistent grade reports, loss of historical academic records, and delayed report card generation. To resolve these issues, the administration is deploying a centralized Database Management System (DBMS). Using the scenario above, write an essay explaining six key significance of a Database Management System (DBMS) in managing institutional data. **(15 Marks)**